0580_w20_ms_43_Q1

Question	Answer	Marks	Partial Marks
1(a)(i)	$5.101[00\ldots] \times 10^8$ final answer	**1**	
1(a)(ii)	361 150 800 oe	**2**	**FT** *their* **(a)(i)** **M1** for $\frac{70.8}{100} \times 510\,100\,000$ or for $\frac{70.8}{100} \times$ *their* **a(i)**
1(b)(i)	6070 oe	**1**	
1(b)(ii)	32 000 oe	**2**	**B1** for figs 32
1(b)(iii)	6.68 or 6.677 …	**2**	**M1** for $\frac{6.41\times10^5}{9.6[0]\times10^6}$ $[\times 100]$ oe
1(b)(iv)	1250 or 1248 to 1249 oe	**2**	**B1** for figs 125 or figs1248 to figs 1249
1(c)(i)	25.1 or 25.08…	**2**	**M1** for $\frac{7.53\left[\times10^9\right]-6.02\left[\times10^9\right]}{6.02\left[\times10^9\right]}$ oe or $\frac{7.53\left[\times10^9\right]}{6.02\left[\times10^9\right]}\times100$
1(c)(ii)	1.33 or 1.325…	**3**	**M2** for $\sqrt[17]{\frac{7.53[\times10^9]}{6.02[\times10^9]}}$ or $\sqrt[17]{1+\frac{\textit{their}\ \textbf{(c)(i)}}{100}}$ or **M1** for $6.02[\times10^9] \times p^{17} = 7.53[\times10^9]$ or $p^{17} = 1+\frac{\textit{their}\ \textbf{(c)(i)}}{100}$